[Tarea 06] Unidad 03-A | Serie de Taylor y Polinomios de Lagrange

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Determine el orden de la mejor aproximación para las siguientes funciones, usando la Serie de Taylor y el Polinomio de Lagrange:

1. $$f(x) = \frac{1}{25x^2+1}, x_0 = 0$$

```
In [1]: import numpy as np
        import matplotlib.pyplot as plt
```

- Serie de Taylor:

$f'(x) = -\frac{50x}{(25x^2+1)^2}$

$f''(x) = -\frac{50}{(25x^2+1)^2} + \frac{5000x^2}{(25x^2+1)^3}$

$f'''(x) = \frac{15000x}{(25x^2+1)^3} - \frac{750000x^3}{(25x^2+1)^4}$

$f^{(4)}(x) = \frac{15000}{(25x^2+1)^3} - \frac{4500000x^2}{(25x^2+1)^4} + \frac{150000000x^4}{(25x^2+1)^5}$

Quedándonos como serie de Taylor de orden 4 con $x_0 = 0$

$$f(0) \approx 1 - 25x^2 + 625x^4$$

así, graficamos cada aproximación

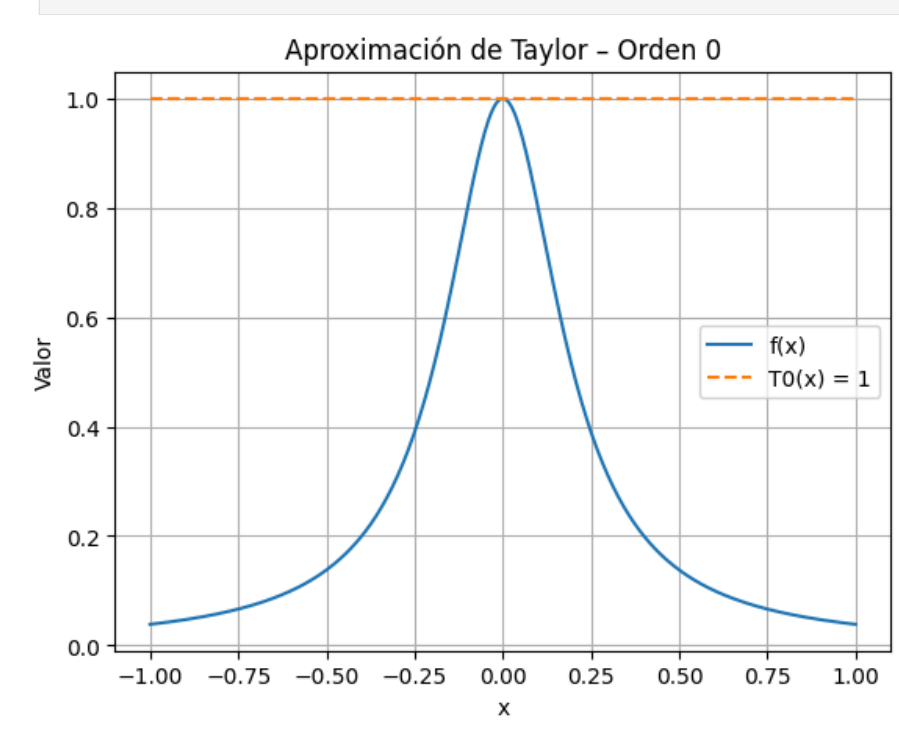
$T_0(x) = 1$

```
In [2]: def f(x):
            return 1/(25*x**2 + 1)

        def T0(x):
            return np.ones_like(x)

        x = np.linspace(-1, 1, 400)

        plt.figure()
        plt.plot(x, f(x), label="f(x)")
        plt.plot(x, T0(x), label="T0(x) = 1", linestyle="--")
        plt.title("Aproximación de Taylor - Orden 0")
        plt.xlabel("x")
        plt.ylabel("Valor")
        plt.legend()
        plt.grid(True)
        plt.show()
```



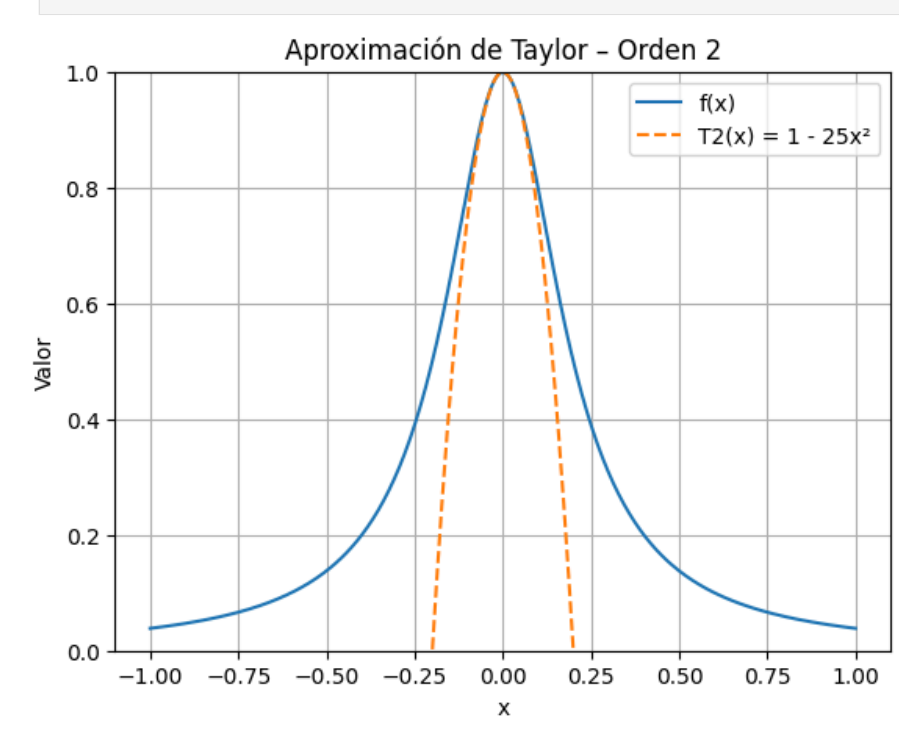
$T_2 = 1 - 25x^2$

```
In [3]: def f(x):
            return 1/(25*x**2 + 1)

        def T2(x):
            return 1 - 25*x**2

        x = np.linspace(-1, 1, 400)

        plt.figure()
        plt.plot(x, f(x), label="f(x)")
        plt.plot(x, T2(x), label="T2(x) = 1 - 25x²", linestyle="--")
        plt.title("Aproximación de Taylor - Orden 2")
        plt.xlabel("x")
        plt.ylabel("Valor")
        plt.ylim (0,1)
        plt.legend()
        plt.grid(True)
        plt.show()
```



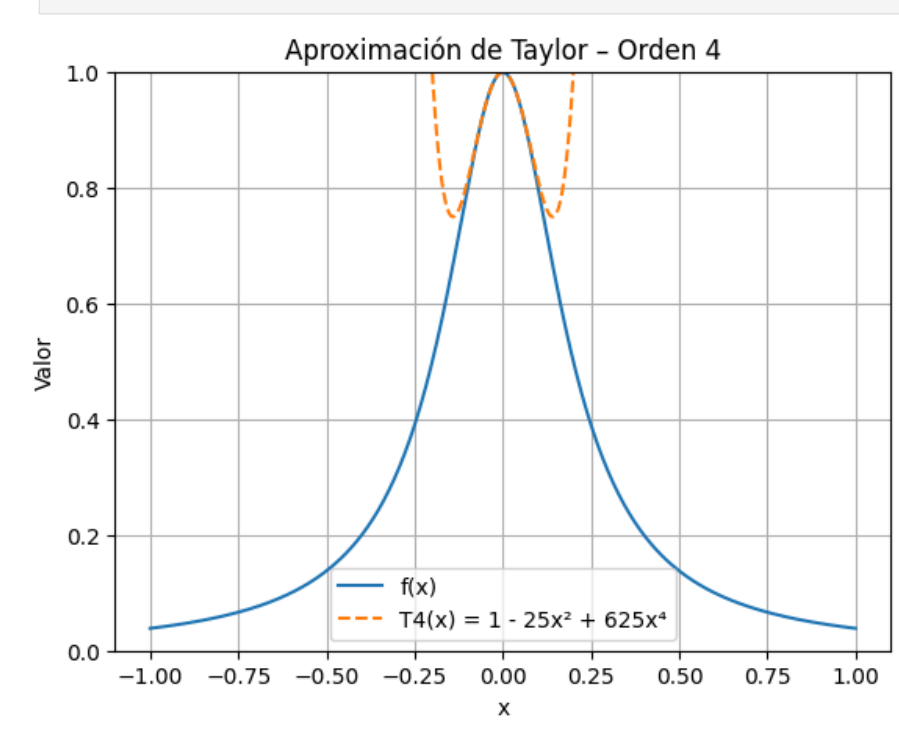
$T_4 = 1 - 25x^2 + 625x^4$

```
In [4]: def f(x):
            return 1/(25*x**2 + 1)

        def T4(x):
            return 1 - 25*x**2 + 625*x**4

        x = np.linspace(-1, 1, 400)

        plt.figure()
        plt.plot(x, f(x), label="f(x)")
        plt.plot(x, T4(x), label="T4(x) = 1 - 25x² + 625x⁴", linestyle="--")
        plt.title("Aproximación de Taylor - Orden 4")
        plt.xlabel("x")
        plt.ylabel("Valor")
        plt.ylim (0,1)
        plt.legend()
        plt.grid(True)
        plt.show()
```



- Polinomio de LaGrange

Como en la serie de Taylor usamos un polinomio de orden 4, para tener un polinomio de orden 4 en el polinomio de LaGrange debemos usar 5 puntos:

$$(0,1)\ \ (0.2,0.5)\ \ (-0.2,0.5)\ \ (0.6,0.1)\ \ (-0.6,0.1)$$

$$P(x) = L_0(x) + 0.5L_1(x) + 0.5L_2(x) + 0.1L_3(x) + 0.1L_4(x)$$

$L_0 = \frac{(x-0.2)(x+0.2)(x-0.6)(x+0.6)}{0.0144}$

$L_1 = -\frac{x(x+0.2)(x-0.6)(x+0.6)}{0.0256}$

$L_2 = -\frac{x(x-0.2)(x-0.6)(x+0.6)}{0.0256}$

$L_3 = \frac{x(x-0.2)(x+0.2)(x+0.6)}{0.2304}$

$L_4 = \frac{x(x-0.2)(x+0.2)(x-0.6)}{0.2304}$

$P(x) = \frac{(x-0.2)(x+0.2)(x-0.6)(x+0.6)}{0.0144} - \frac{1}{2}(\frac{x(x+0.2)(x-0.6)(x+0.6)}{0.0256}) - \frac{1}{2}(\frac{x(x-0.2)(x-0.6)(x+0.6)}{0.0256}) + \frac{1}{10}(\frac{x(x-0.2)(x+0.2)(x+0.6)}{0.2304}) + \frac{1}{10}(\frac{x(x-0.2)(x+0.2)(x-0.6)}{0.2304})$

$$P(x) = 31.25x^4 - 13.75x^2 + 1$$

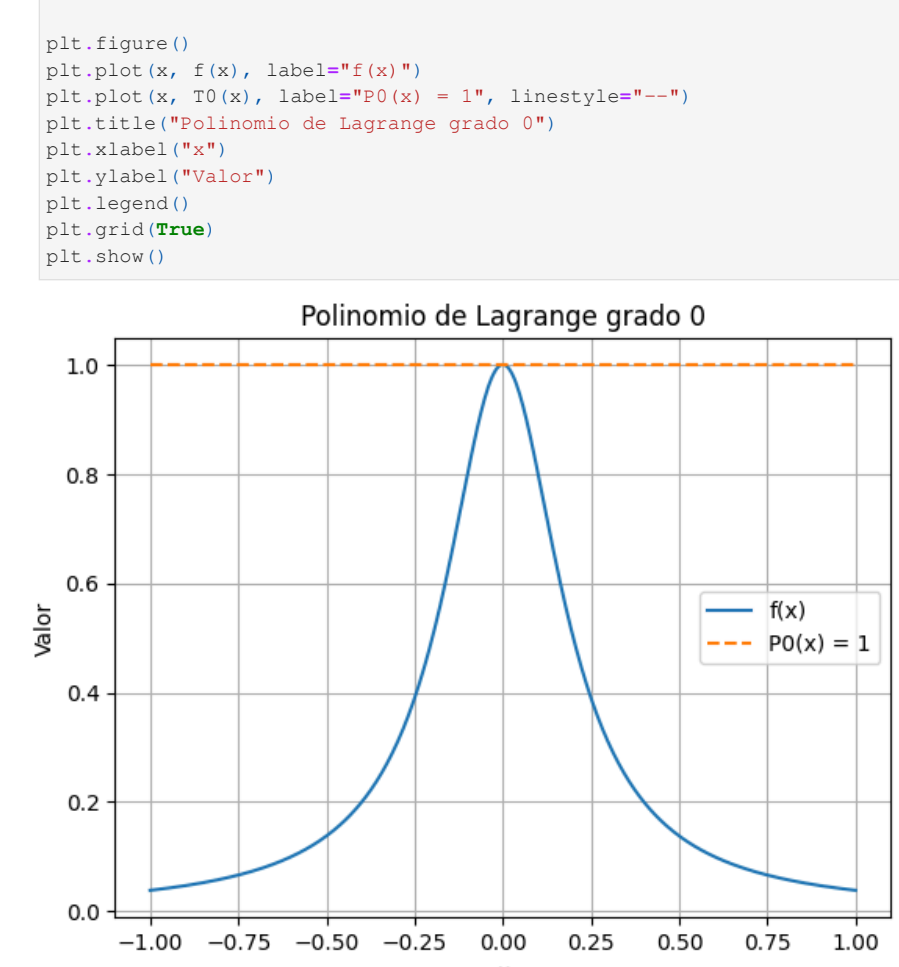
$P_0(x) = 1$

```
In [5]: def f(x):
            return 1/(25*x**2 + 1)

        def T0(x):
            return np.ones_like(x)

        x = np.linspace(-1, 1, 400)

        plt.figure()
        plt.plot(x, f(x), label="f(x)")
        plt.plot(x, T0(x), label="P0(x) = 1", linestyle="--")
        plt.title("Polinomio de Lagrange grado 0")
        plt.xlabel("x")
        plt.ylabel("Valor")
        plt.legend()
        plt.grid(True)
        plt.show()
```



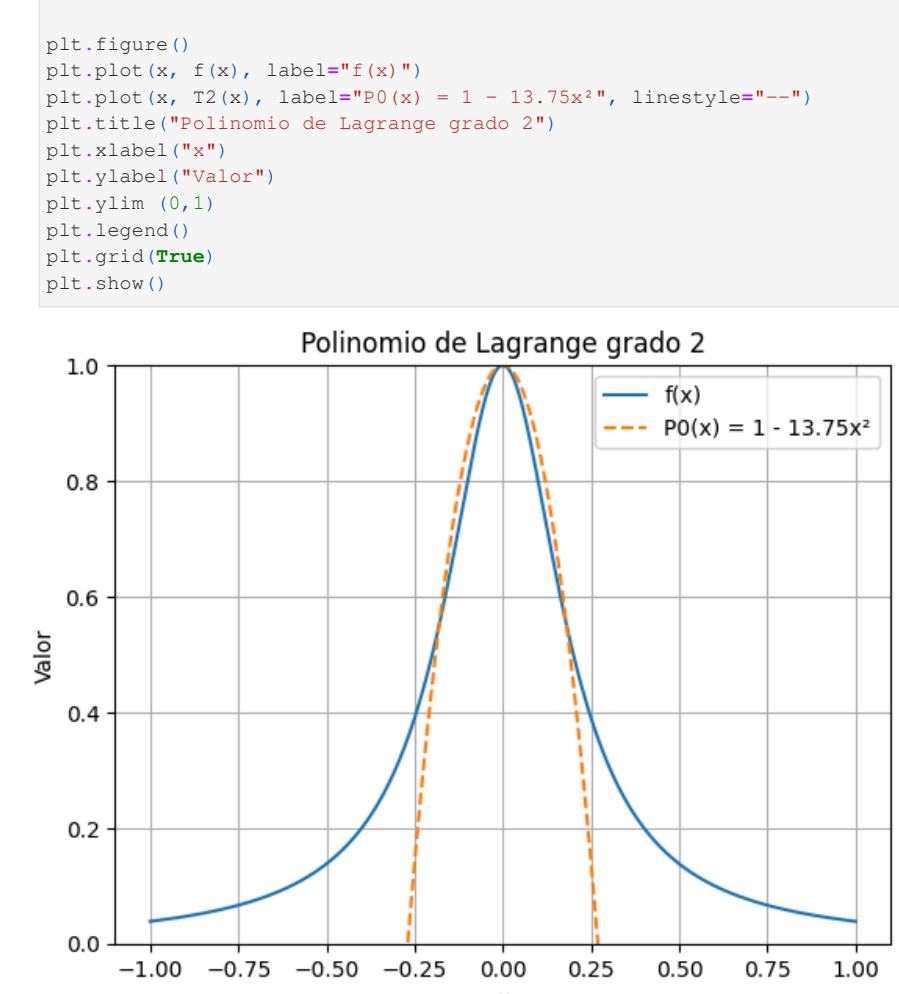
$P_2(x) = 1 - 13.75x^2$

```
In [6]: def f(x):
            return 1/(25*x**2 + 1)

        def T2(x):
            return 1 - 13.75*x**2

        x = np.linspace(-1, 1, 400)

        plt.figure()
        plt.plot(x, f(x), label="f(x)")
        plt.plot(x, T2(x), label="P0(x) = 1 - 13.75x²", linestyle="--")
        plt.title("Polinomio de Lagrange grado 2")
        plt.xlabel("x")
        plt.ylabel("Valor")
        plt.ylim (0,1)
        plt.legend()
        plt.grid(True)
        plt.show()
```



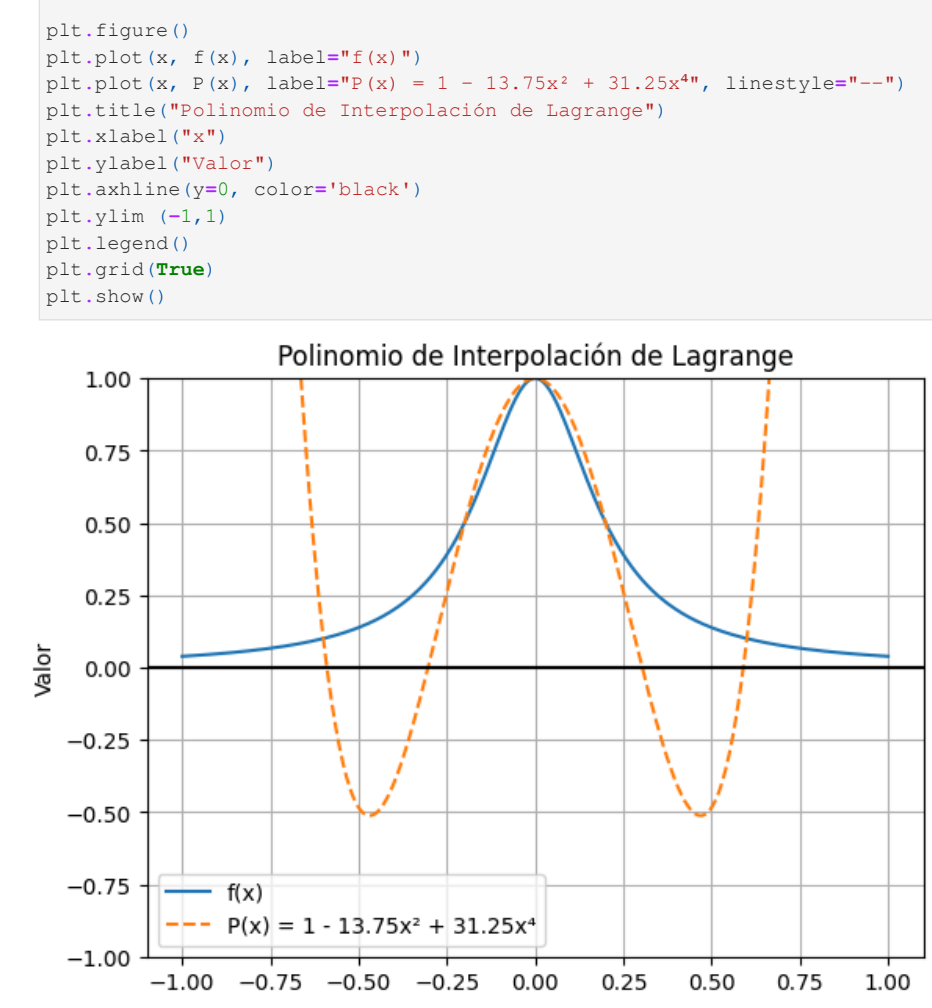
$P_4(x) = 1 - 13.75x^2 + 31.25x^4$

```
In [7]: def f(x):
            return 1/(25*x**2 + 1)

        def P(x):
            return 31.25*x**4 - 13.75*x**2 + 1

        x = np.linspace(-1, 1, 400)

        plt.figure()
        plt.plot(x, f(x), label="f(x)")
        plt.plot(x, P(x), label="P(x) = 1 - 13.75x² + 31.25x⁴", linestyle="--")
        plt.title("Polinomio de Interpolación de Lagrange")
        plt.xlabel("x")
        plt.ylabel("Valor")
        plt.axhline(y=0, color='black')
        plt.ylim (-1,1)
        plt.legend()
        plt.grid(True)
        plt.show()
```



El orden de la mejor aproximación es de 4

2. $$f(x) = arctanx, x_0 = 1$$

- Serie de Taylor

$f'(x) = \frac{1}{1+x^2}$

$f''(x) = -\frac{2x}{(1+x^2)^2}$

$f'''(x) = -\frac{2}{(1+x^2)^2} + \frac{8x^2}{(1+x^2)^3}$

$f^{(4)}(x) = \frac{24x}{(1+x^2)^3} - \frac{48x^3}{(1+x^2)^4}$

$f^{(5)}(x) = \frac{24}{(1+x^2)^3} - \frac{288x^2}{(1+x^2)^4} + \frac{384x^4}{(1+x^2)^5}$

Quedándonos como serie de Taylor con orden 5:

$$f(1) \approx \frac{\pi}{4} + \frac{1}{2}(x-1) - \frac{1}{4}(x-1)^2 + \frac{1}{12}(x-1)^3 - \frac{1}{40}(x-1)^5$$

Así, graficamos cada aproximación

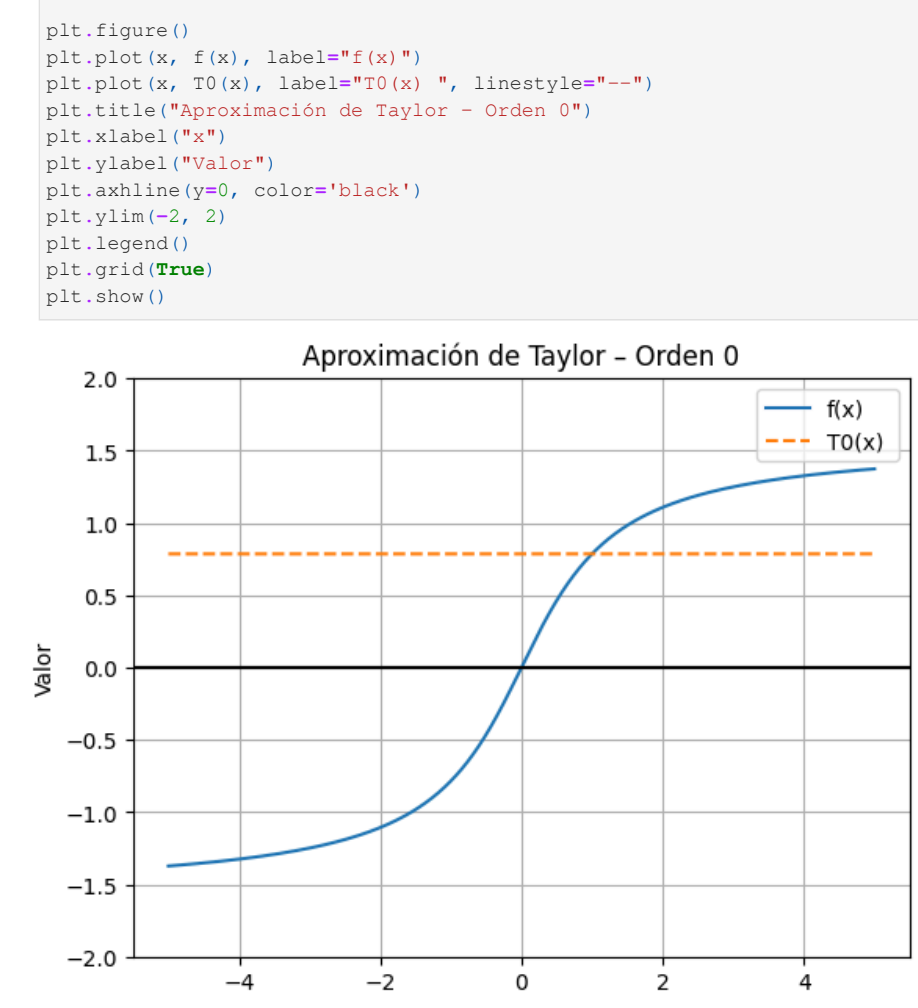
$T_0(x) = \frac{\pi}{4}$

```
In [8]: def f(x):
            return np.arctan(x)

        def T0(x):
            return np.ones_like(x) * np.pi/4

        x = np.linspace(-5, 5, 2000)

        plt.figure()
        plt.plot(x, f(x), label="f(x)")
        plt.plot(x, T0(x), label="T0(x) ", linestyle="--")
        plt.title("Aproximación de Taylor - Orden 0")
        plt.xlabel("x")
        plt.ylabel("Valor")
        plt.axhline(y=0, color='black')
        plt.ylim(-2, 2)
        plt.legend()
        plt.grid(True)
        plt.show()
```



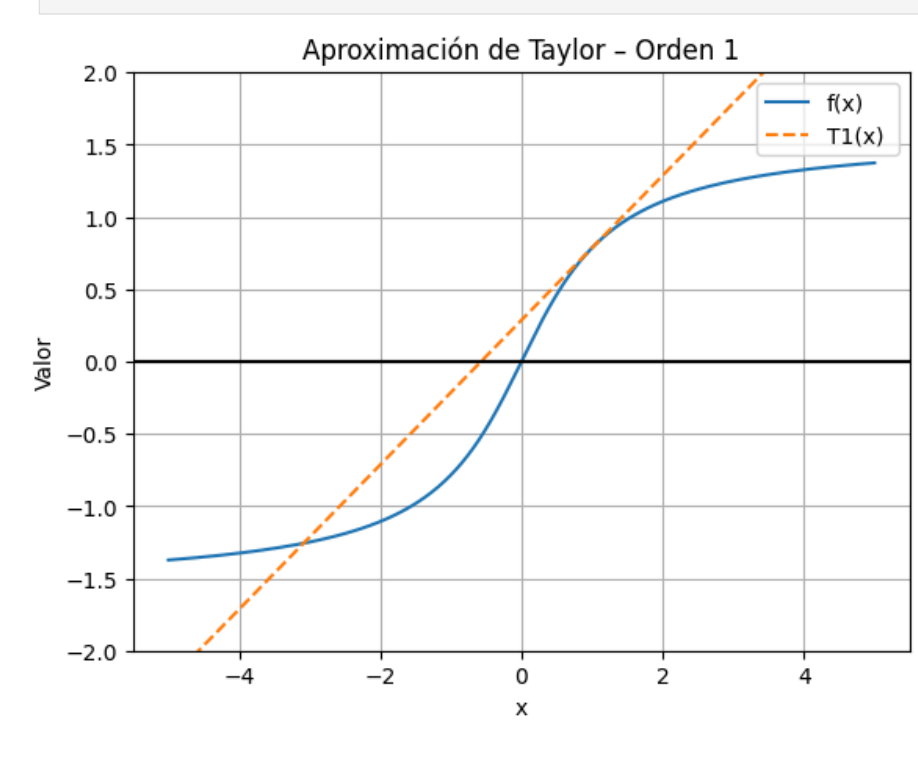
$T_1(x) = \frac{\pi}{4} + \frac{1}{2}(x-1)$

```
In [9]: def f(x):
            return np.arctan(x)

        def T0(x):
            return  np.pi/4 + 1/2*(x-1)

        x = np.linspace(-5, 5, 2000)

        plt.figure()
        plt.plot(x, f(x), label="f(x)")
        plt.plot(x, T0(x), label="T1(x) ", linestyle="--")
        plt.title("Aproximación de Taylor - Orden 1")
        plt.xlabel("x")
        plt.ylabel("Valor")
        plt.ylim(-2, 2)
        plt.axhline(y=0, color='black')
        plt.legend()
        plt.grid(True)
        plt.show()
```



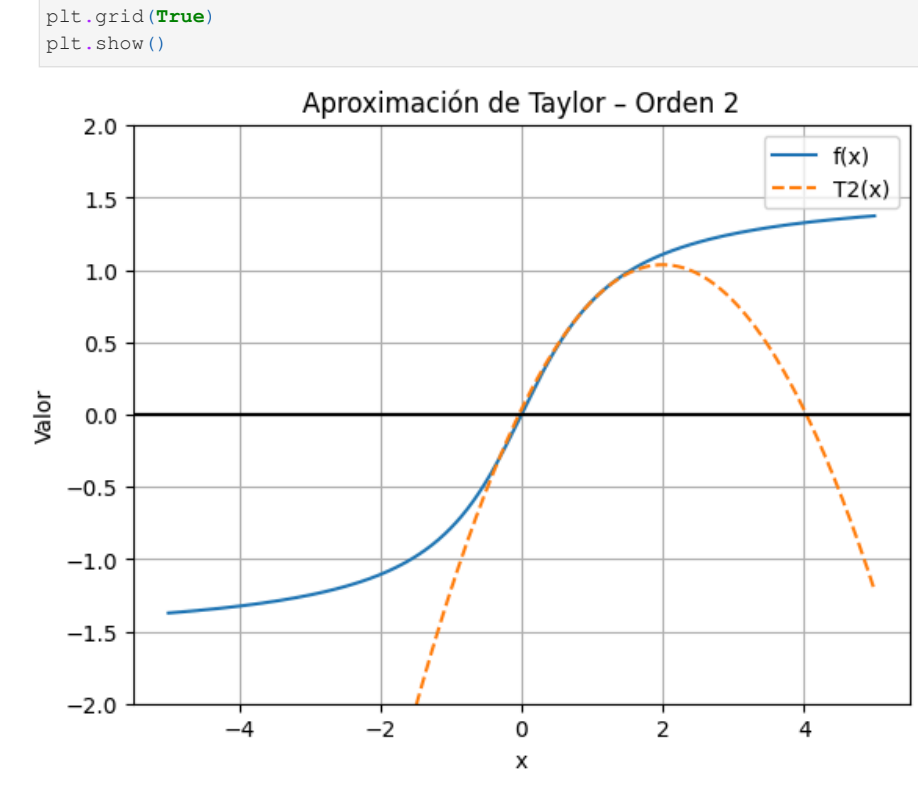
$T_2(x) = \frac{\pi}{4} + \frac{1}{2}(x-1) - \frac{1}{4}(x-1)^2$

```
In [10]: def f(x):
             return np.arctan(x)

         def T0(x):
             return  np.pi/4 + 1/2*(x-1) - 1/4*(x-1)**2

         x = np.linspace(-5, 5, 2000)

         plt.figure()
         plt.plot(x, f(x), label="f(x)")
         plt.plot(x, T0(x), label="T2(x)", linestyle="--")
         plt.title("Aproximación de Taylor - Orden 2")
         plt.xlabel("x")
         plt.ylabel("Valor")
         plt.ylim(-2, 2)
         plt.axhline(y=0, color='black')
         plt.legend()
         plt.grid(True)
         plt.show()
```



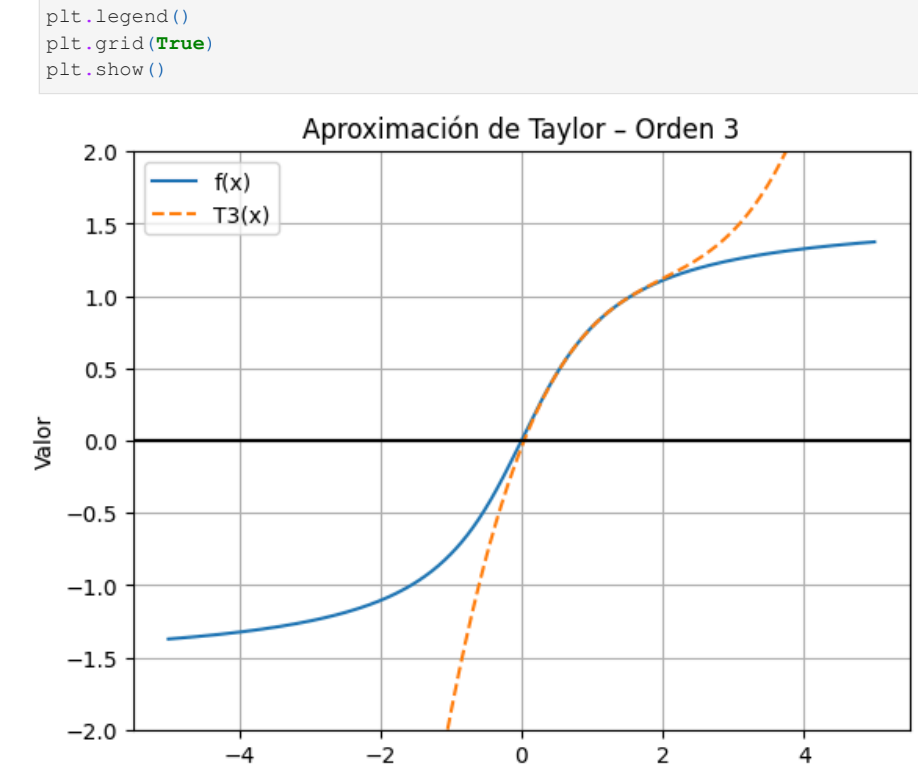
$T_3(x) = \frac{\pi}{4} + \frac{1}{2}(x-1) - \frac{1}{4}(x-1)^2 + \frac{1}{12}(x-1)^3$

```
In [11]: def f(x):
             return np.arctan(x)

         def T0(x):
             return  np.pi/4 + 1/2*(x-1) - 1/4*(x-1)**2 + 1/12*(x-1)**3

         x = np.linspace(-5, 5, 2000)

         plt.figure()
         plt.plot(x, f(x), label="f(x)")
         plt.plot(x, T0(x), label="T3(x)", linestyle="--")
         plt.title("Aproximación de Taylor - Orden 3")
         plt.xlabel("x")
         plt.ylabel("Valor")
         plt.ylim(-2, 2)
         plt.axhline(y=0, color='black')
         plt.legend()
         plt.grid(True)
         plt.show()
```



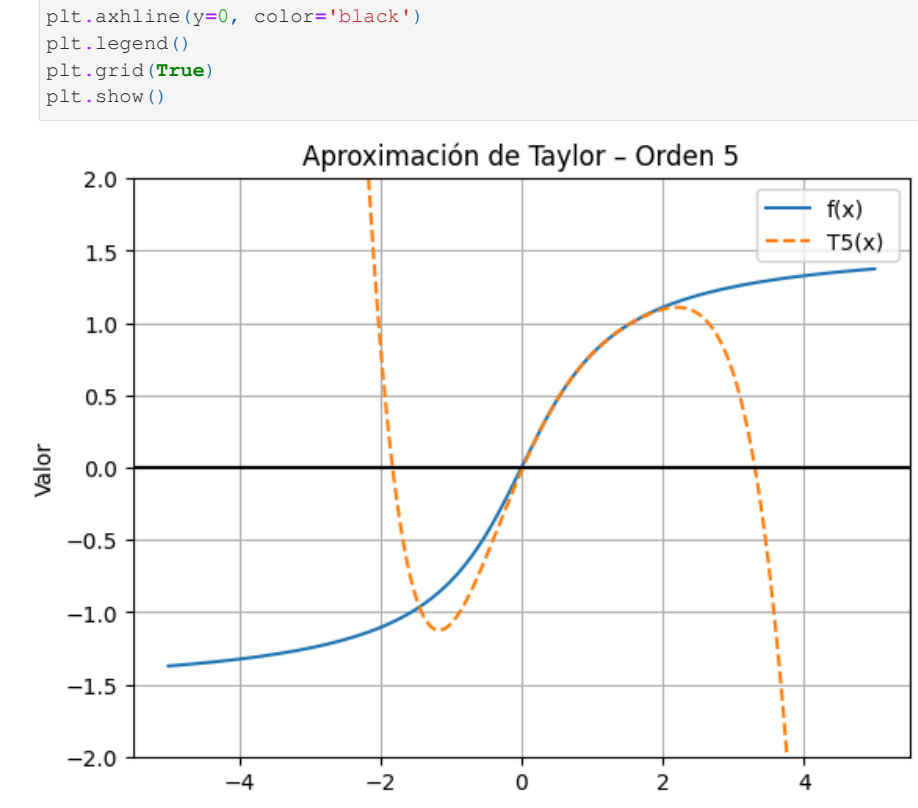
$T_5(x) = \frac{\pi}{4} + \frac{1}{2}(x-1) - \frac{1}{4}(x-1)^2 + \frac{1}{12}(x-1)^3 - \frac{1}{40}(x-1)^5$

```
In [12]: def f(x):
             return np.arctan(x)

         def T0(x):
             return  np.pi/4 + 1/2*(x-1) - 1/4*(x-1)**2 + 1/12*(x-1)**3 - 1/40*(x-1)**5

         x = np.linspace(-5, 5, 2000)

         plt.figure()
         plt.plot(x, f(x), label="f(x)")
         plt.plot(x, T0(x), label="T5(x) ", linestyle="--")
         plt.title("Aproximación de Taylor - Orden 5")
         plt.xlabel("x")
         plt.ylabel("Valor")
         plt.ylim(-2, 2)
         plt.axhline(y=0, color='black')
         plt.legend()
         plt.grid(True)
         plt.show()
```



- Polinomio de LaGrange

Como en la serie de Taylor usamos un polinomio de orden 5, para tener un polinomio de orden 5 en el polinomio de LaGrange debemos usar 6 puntos:

$$(-1,-\frac{\pi}{4})\ \ (0,0)\ \ (0.5,0.4636)\ \ (1,\frac{\pi}{4})\ \ (1.5,0.9828)\ \ (2,1.1071)$$

$L_0(x) = -\frac{2}{45}x^5 + \frac{2}{9}x^4 - \frac{7}{18}x^3 + \frac{5}{18}x^2 - \frac{1}{15}x$

$L_1(x) = \frac{2}{3}x^5 - \frac{8}{3}x^4 + \frac{5}{2}x^3 + \frac{5}{3}x^2 - \frac{19}{6}x + 1$

$L_2(x) = -\frac{16}{9}x^5 + \frac{56}{9}x^4 - \frac{32}{9}x^3 - \frac{56}{9}x^2 + \frac{16}{3}x$

$L_3(x) = 2x^5 - 6x^4 + \frac{3}{2}x^3 + \frac{13}{2}x^2 - 3x$

$L_4(x) = -\frac{16}{15}x^5 + \frac{8}{3}x^4 - \frac{8}{3}x^2 + \frac{16}{15}x$

$L_2(x) = \frac{2}{9}x^5 - \frac{4}{9}x^4 - \frac{1}{18}x^3 + \frac{4}{9}x^2 - \frac{1}{16}x$

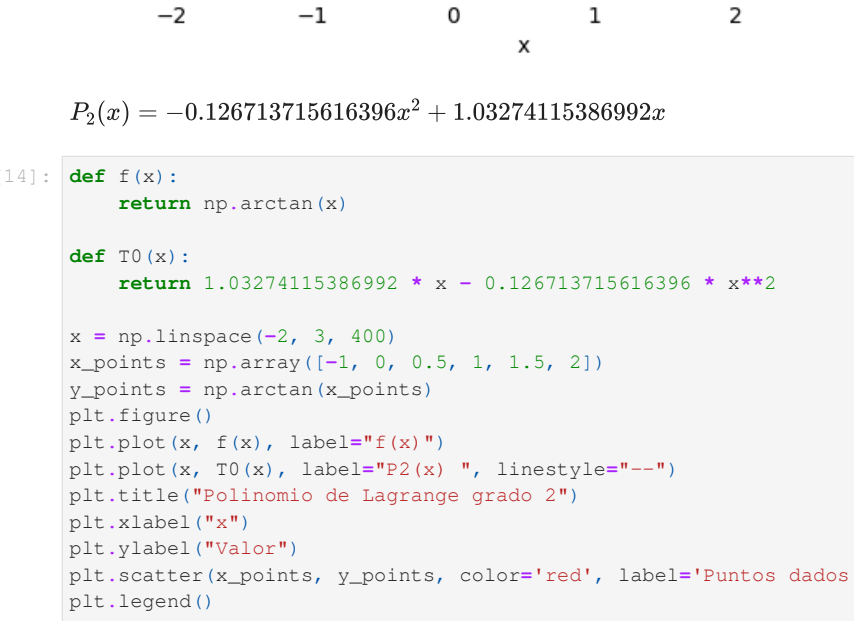
$$P(x) = -0.0208397605651139x^5 + 0.126713715616397x^4 - 0.226503229907357x^3 - 0.126713715616396x^2 + 1.03274115386992x$$

$P_1(x) = 1.03274115386992x$

```
In [13]: def f(x):
             return np.arctan(x)

         def T0(x):
             return 1.03274115386992 * x

         x = np.linspace(-2, 3, 400)
         x_points = np.array([-1, 0, 0.5, 1, 1.5, 2])
         y_points = np.arctan(x_points)
         plt.figure()
         plt.plot(x, f(x), label="f(x)")
         plt.plot(x, T0(x), label="P1(x) ", linestyle="--")
         plt.title("Polinomio de Lagrange grado 1")
         plt.xlabel("x")
         plt.ylabel("Valor")
         plt.scatter(x_points, y_points, color='red', label='Puntos dados')
         plt.legend()
         plt.grid(True)
         plt.show()
```

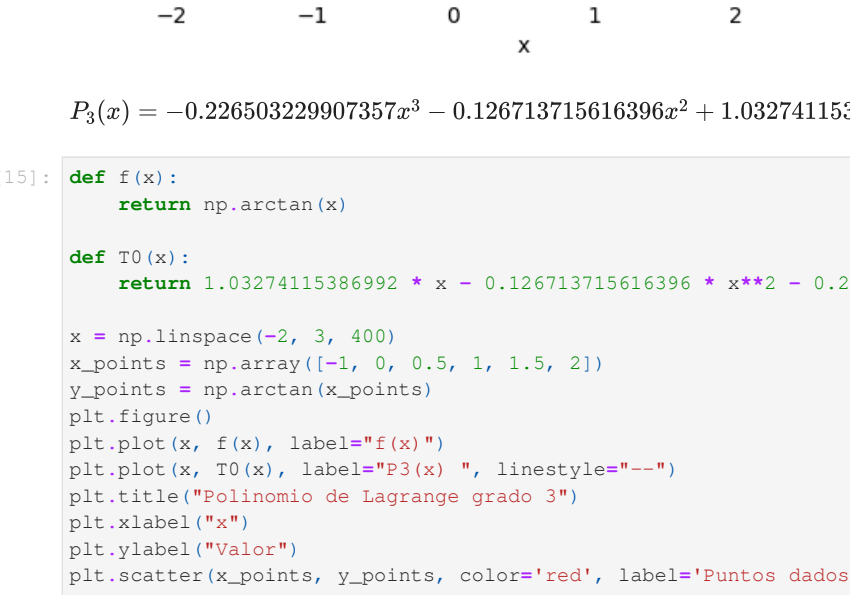


$P_2(x) = -0.126713715616396x^2 + 1.03274115386992x$

```
In [14]: def f(x):
             return np.arctan(x)

         def T0(x):
             return 1.03274115386992 * x - 0.126713715616396 * x**2

         x = np.linspace(-2, 3, 400)
         x_points = np.array([-1, 0, 0.5, 1, 1.5, 2])
         y_points = np.arctan(x_points)
         plt.figure()
         plt.plot(x, f(x), label="f(x)")
         plt.plot(x, T0(x), label="P2(x) ", linestyle="--")
         plt.title("Polinomio de Lagrange grado 2")
         plt.xlabel("x")
         plt.ylabel("Valor")
         plt.scatter(x_points, y_points, color='red', label='Puntos dados')
         plt.legend()
         plt.grid(True)
         plt.show()
```

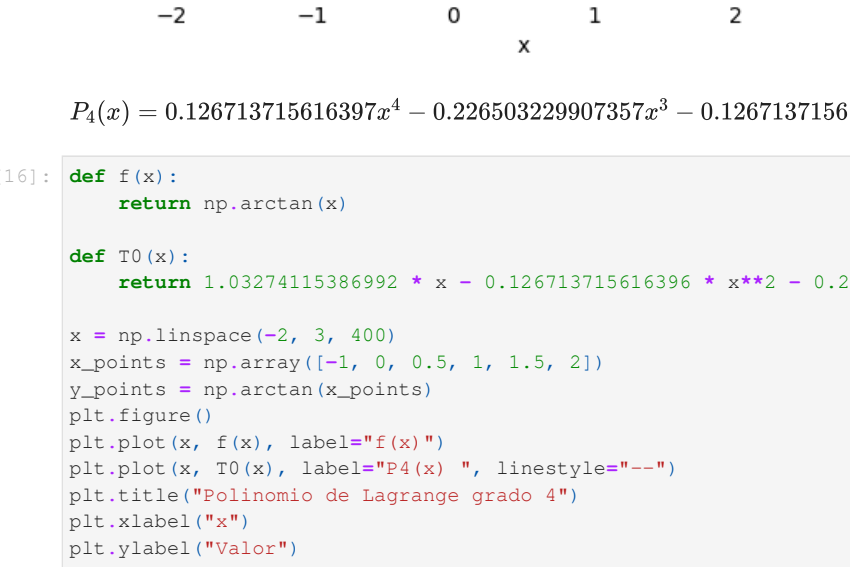


$P_3(x) = -0.226503229907357x^3 - 0.126713715616396x^2 + 1.03274115386992x$

```
In [15]: def f(x):
             return np.arctan(x)

         def T0(x):
             return 1.03274115386992 * x - 0.126713715616396 * x**2 - 0.226503229907357 * x**3

         x = np.linspace(-2, 3, 400)
         x_points = np.array([-1, 0, 0.5, 1, 1.5, 2])
         y_points = np.arctan(x_points)
         plt.figure()
         plt.plot(x, f(x), label="f(x)")
         plt.plot(x, T0(x), label="P3(x) ", linestyle="--")
         plt.title("Polinomio de Lagrange grado 3")
         plt.xlabel("x")
         plt.ylabel("Valor")
         plt.scatter(x_points, y_points, color='red', label='Puntos dados')
         plt.legend()
         plt.grid(True)
         plt.show()
```

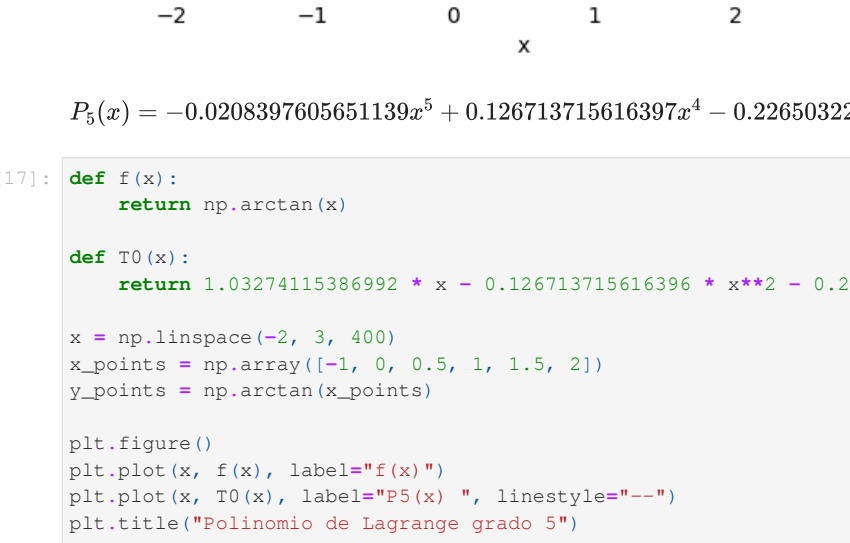


$P_4(x) = 0.126713715616397x^4 - 0.226503229907357x^3 - 0.126713715616396x^2 + 1.03274115386992x$

```
In [16]: def f(x):
             return np.arctan(x)

         def T0(x):
             return 1.03274115386992 * x - 0.126713715616396 * x**2 - 0.226503229907357 * x**3 + 0.126713715616397 * x**4

         x = np.linspace(-2, 3, 400)
         x_points = np.array([-1, 0, 0.5, 1, 1.5, 2])
         y_points = np.arctan(x_points)
         plt.figure()
         plt.plot(x, f(x), label="f(x)")
         plt.plot(x, T0(x), label="P4(x) ", linestyle="--")
         plt.title("Polinomio de Lagrange grado 4")
         plt.xlabel("x")
         plt.ylabel("Valor")
         plt.scatter(x_points, y_points, color='red', label='Puntos dados')
         plt.legend()
         plt.grid(True)
         plt.show()
```



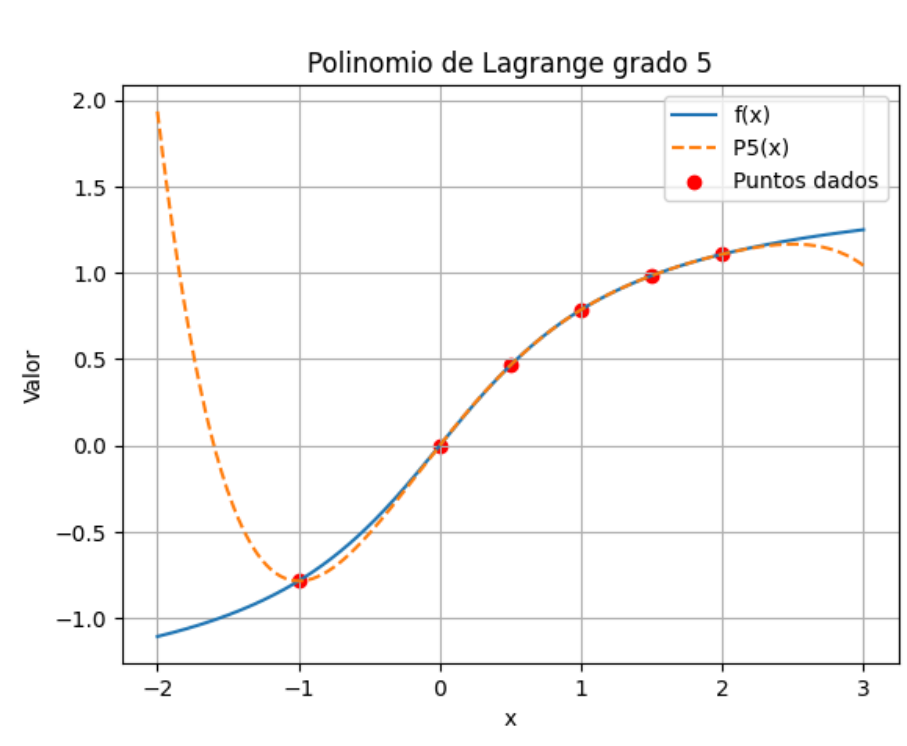
$P_5(x) = -0.0208397605651139x^5 + 0.126713715616397x^4 - 0.226503229907357x^3 - 0.126713715616396x^2 + 1.03274115386992x$

```
In [17]: def f(x):
             return np.arctan(x)

         def T0(x):
             return 1.03274115386992 * x - 0.126713715616396 * x**2 - 0.226503229907357 * x**3 + 0.126713715616397 * x**4 - 0.0208397605651139 * x**5

         x = np.linspace(-2, 3, 400)
         x_points = np.array([-1, 0, 0.5, 1, 1.5, 2])
         y_points = np.arctan(x_points)

         plt.figure()
         plt.plot(x, f(x), label="f(x)")
         plt.plot(x, T0(x), label="P5(x) ", linestyle="--")
         plt.title("Polinomio de Lagrange grado 5")
         plt.xlabel("x")
         plt.ylabel("Valor")
         plt.scatter(x_points, y_points, color='red', label='Puntos dados')
         plt.legend()
         plt.grid(True)
         plt.show()
```



El mejor orden de la aproximación de 5